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OPERATING SYSTEM (ADVANCED)WK6

1.SCENARIO ANALYSIS:

Device Management Scenario: Server Driver Management

A company operates a server that manages its employee database, email services, and internal applications. The server depends on different hardware devices, such as network adapters, storage controllers, printers, and backup devices. Each device requires a suitable driver so that the operating system can communicate with and control the hardware correctly. One day, the server administrator installs a new network adapter, but the correct driver is not installed. As a result, the server cannot connect properly to the company network, causing employees to lose access to important services. The administrator uses device management tools to identify the problem and checks the device status for driver errors. The correct driver is downloaded from the hardware manufacturer's website and installed. After restarting the server, the network adapter works normally. This scenario demonstrates the importance of proper device management, regular driver updates, compatibility checking, and monitoring to maintain reliable server performance and reduce system downtime.

2.CONCEPT RESEARCH:

Concept Research: DMA

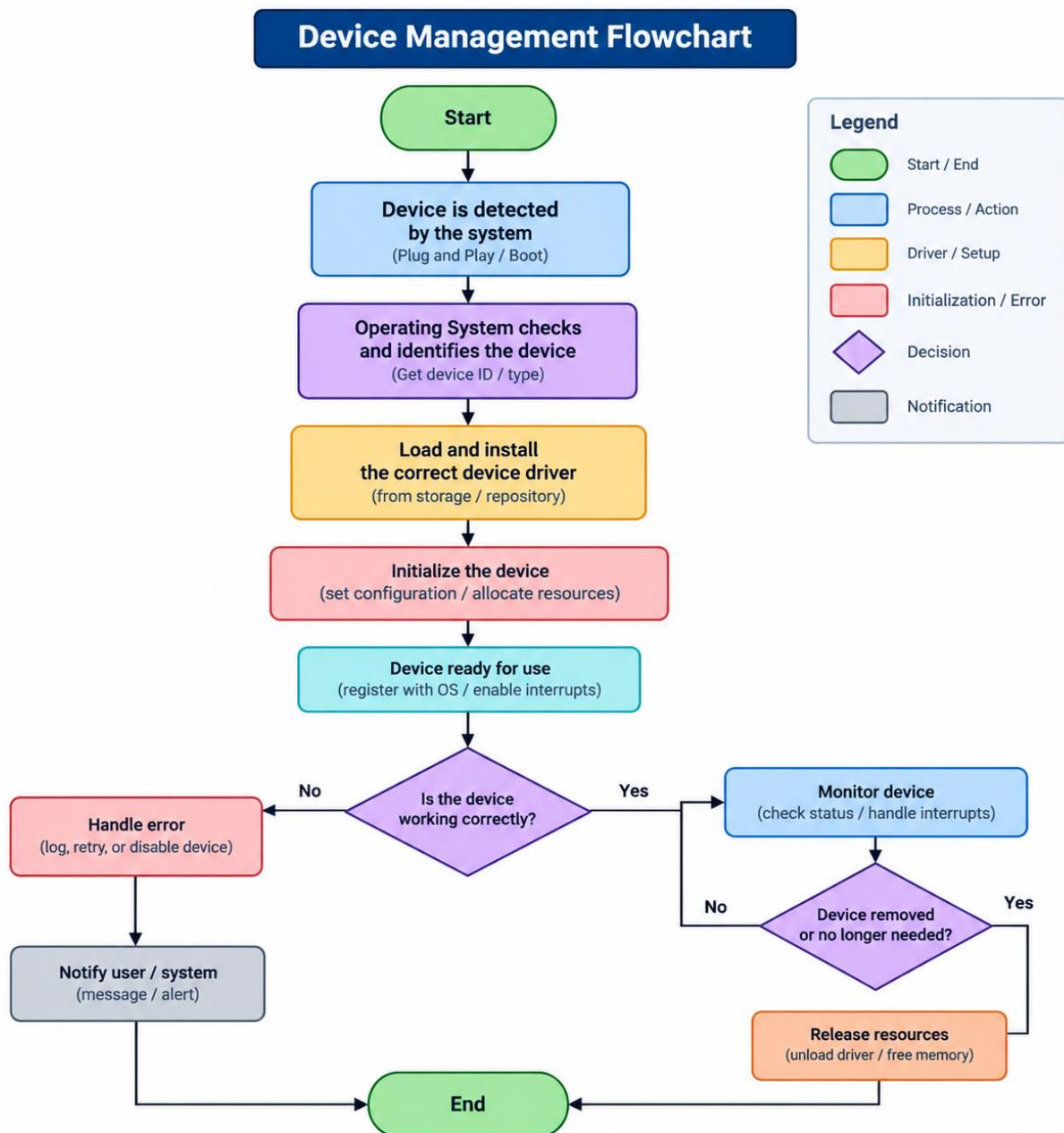
Direct Memory Access (DMA) is a computer system technique that allows hardware devices to transfer data directly to or from main memory without requiring the CPU to handle every byte of data. A DMA controller manages the transfer by receiving instructions from the CPU, accessing memory, and communicating with the device. This reduces CPU workload and improves overall system performance. DMA is commonly used for high-speed devices such as disk drives, network cards, sound cards, and graphics devices. Once the transfer is completed, the DMA controller can notify the CPU through an interrupt. Therefore, DMA provides faster and more efficient data transfer between devices and memory.

3.TOOL PRACTICE:

I simulated an interrupt in QEMU to understand how a computer system responds to an external event. During the simulation, a device generated an interrupt request, causing the CPU to temporarily stop its current task and execute the appropriate interrupt service routine (ISR). After the ISR completed, the CPU returned to the interrupted program and continued execution. This practice helped me understand the communication between hardware and the operating system. I learned that interrupts allow devices such as keyboards, timers, and network cards to notify the CPU when

they require attention. Overall, the QEMU simulation demonstrated how interrupts improve system efficiency by allowing the CPU to respond to important events without constantly checking every device.

4.DIAGRAM:



The device management flowchart shows how an operating system manages a hardware device from detection to removal. First, the system detects the device and identifies its type. Next, the correct device driver is loaded and installed so the operating system can communicate with the hardware. The device is then initialized by

configuring settings and allocating the required resources. After initialization, the device becomes ready for use and its operation is monitored. A decision checks whether the device is working correctly. If an error occurs, the system handles the error and notifies the user. When the device is no longer needed, its resources are released before the process ends.